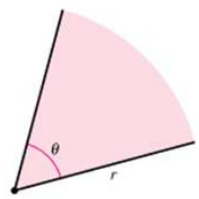
Topic 9.8: Finding the Area of a Polar Region or the Area Bounded by a Single Polar Curve

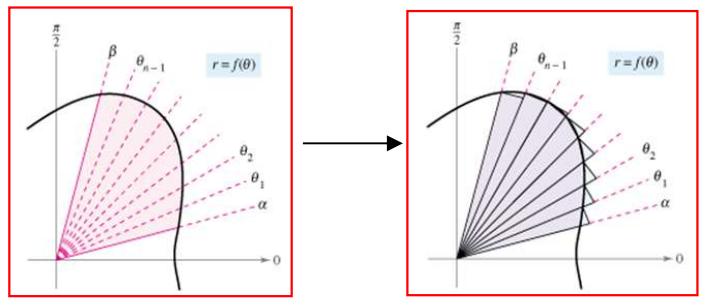
Learning Objective CHA-5.D: Calculate areas of regions defined by polar curves by using definite integrals.

Area of a Polar Region



The area of a sector of a circle is $A = \frac{1}{2}\theta r^2$.

The development of a formula for the area of a polar region parallels that for the area of the region on the rectangular coordinate system. The primary difference is that in polar, we use **sectors of circles** instead of rectangles to partition off our region.



Consider the function given by $r = f(\theta)$, where f is continuous and nonnegative on the interval given by $\alpha \le \theta \le \beta$. The region bounded by f and the radial lines $\theta = \alpha$ and $\theta = \beta$ is what we wish to find.

If we partition $[\alpha, \beta]$ into n subintervals where $\alpha = \theta_0 < \theta_1 < \theta_2 < \cdots < \theta_{n-1} < \theta_n = \beta$

And let the radius of the ith sector be $f(\theta_i)$ and the central angle of the ith sector be $\frac{\beta - \alpha}{n} = \Delta\theta$,

Then our area can be approximated as $A \approx \sum_{i=1}^{n} \left(\frac{1}{2}\right) \Delta\theta \left[f(\theta_i)\right]^2$.

Taking the limit as $n \to \infty$ produces

$$A = \lim_{n\to\infty} \frac{1}{2} \sum_{i=1}^{n} \left[f(\theta_i)\right]^2 \Delta\theta$$

$$= \frac{1}{2} \int_{\alpha}^{\beta} \left[f(\theta)\right]^2 d\theta$$

THEOREM 9.8: AREA IN POLAR COORDINATES

If f is continuous and nonnegative on the interval $[\alpha, \beta]$, $0 < \beta - \alpha < 2\pi$ then the area of the region bounded by the graph of $r = f(\theta)$ between the radial lines $\theta = \alpha$ and $\theta = \beta$ is given by

$$A = \frac{1}{2} \int_{\alpha}^{\beta} [f(\theta)]^2 \, d\theta$$

<u>Example 1</u>: <u>Finding the Area of a Polar Region</u>.

Find the area of one petal of the rose curve given by $r = 3\cos 3\theta$.

$$A = 2 \cdot \frac{1}{2} \int_0^{\pi/6} (3\cos 3\theta)^2 \, d\theta$$

$$A = 9 \int_0^{\pi/6} \cos^2 3\theta \, d\theta$$

$$= \frac{9}{2} \int_0^{\pi/6} (1 + \cos 6\theta) \, d\theta$$

$$\cos^2\theta = \frac{1}{2}(1 + \cos 2\theta)$$

$$A = \frac{9}{2}\left[\theta + \frac{1}{6}\sin 6\theta\right]_0^{\pi/6}$$

$$A = \frac{9}{2}\left(\frac{\pi}{6} + 0 - (0 + 0)\right)$$

$$= \frac{3\pi}{4}$$

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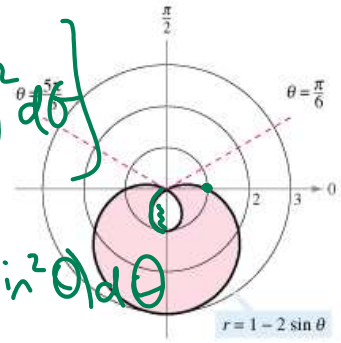
Scan the QR Code above to watch a video covering Example 1

Example 2: Finding the Area Bounded by a Single Curve $\theta = 0 \Rightarrow r = 1$ $\theta = \frac{\pi}{6}$

Find the area of the region lying between the inner and outer loops of the limaçon $r = 1 - 2\sin\theta$. Use your TI-Nspire to sketch a graph of the curve to confirm the boundaries of integration that are shown in the figure below. Do not use your TI-Nspire to evaluate.



$$A = 2\left[\frac{1}{2}\int_{\frac{5\pi}{6}}^{\frac{3\pi}{2}}(1-2\sin\theta)^2\,d\theta - \frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{\pi}{2}}(1-2\sin\theta)^2\,d\theta\right]$$

$$A = \int_{\frac{5\pi}{6}}^{\frac{3\pi}{2}}(1-4\sin\theta+4\sin^2\theta)\,d\theta - \int_{\frac{\pi}{6}}^{\frac{\pi}{2}}(1-4\sin\theta+4\sin^2\theta)\,d\theta$$

$$A = \int_{\frac{5\pi}{6}}^{\frac{3\pi}{2}}(3-4\sin\theta-2\cos 2\theta)\,d\theta - \int_{\frac{\pi}{6}}^{\frac{\pi}{2}}(3-4\sin\theta-2\cos 2\theta)\,d\theta$$

$$A = \left[3\theta+4\cos\theta-\sin 2\theta\right]_{\frac{5\pi}{6}}^{\frac{3\pi}{2}} - \left[3\theta+4\cos\theta-\sin 2\theta\right]_{\frac{\pi}{6}}^{\frac{\pi}{2}}$$

$$A\left(\frac{9\pi}{2}-\left(\frac{5\pi}{2}-2\sqrt{3}+\frac{\sqrt{3}}{2}\right)\right)-\left(\frac{3\pi}{2}-\left(\frac{\pi}{2}+2\sqrt{3}-\frac{\sqrt{3}}{2}\right)\right)$$

$$A = \pi + 3\sqrt{3}$$

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Points of Intersection of Polar Graphs

ACTIVITY: **1.** Locate the points of intersection of the polar graphs $r = 1 - 2\cos\theta$ and $r = 1$ by solving algebraically.

$r = -1$

$(1, \pi)$

$1 = 1 - 2\cos\theta$

$\cos\theta = 0$

$\theta = \frac{\pi}{2}$ $\quad (1, \frac{\pi}{2})$

$\theta = \frac{3\pi}{2}$ $\quad (1, \frac{3\pi}{2})$

2. Use your graphing calculator to sketch both graphs. What do you notice about the points of intersection?

Missing point, because we should consider $\pm$ values for r.

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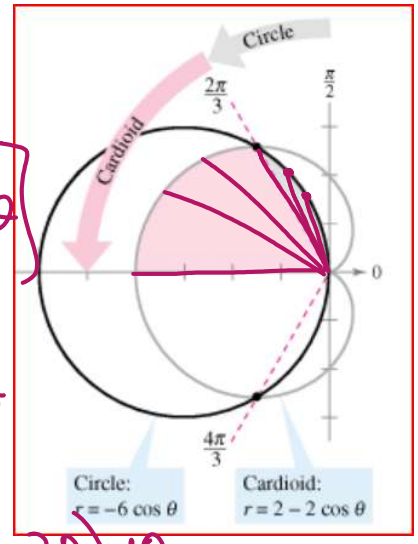
Scan the QR Code above to watch a video covering this activity.

Topic 9.9: Finding the Area of the Region Bounded by Two Polar Curves

Learning Objective CHA-5.D: Calculate areas of regions defined by polar curves by using definite integrals.

<u>Example 1</u>: Finding the Area of a Region Between Two Curves

Find the area of the region common to the two regions bounded by the following curves $\begin{matrix} r = -6\cos\theta \\ r = 2 - 2\cos\theta \end{matrix}$. Note: The shaded region in the picture only shows half of the region whose area you are trying to find.



$$A = 2\left[\frac{1}{2}\int_{\pi/2}^{2\pi/3}(-6\cos\theta)^2\,d\theta + \frac{1}{2}\int_{\frac{2\pi}{3}}^{\pi}(2-2\cos\theta)^2\,d\theta\right]$$

$$A = \int_{\pi/2}^{2\pi/3} 36\cos^2\theta\,d\theta + \int_{\frac{2\pi}{3}}^{\pi}(4 - 8\cos\theta + 4\cos^2\theta)\,d\theta$$

$$A = \int_{\pi/2}^{2\pi/3} 18(1+\cos 2\theta)\,d\theta + \int_{\frac{2\pi}{3}}^{\pi}(6 - 8\cos\theta + 2\cos 2\theta)\,d\theta$$

$$A = 18\left[\theta + \frac{1}{2}\sin 2\theta\right]_{\pi/2}^{2\pi/3} + \left[6\theta - 8\sin\theta + \sin 2\theta\right]_{\frac{2\pi}{3}}^{\pi}$$

$$A = 18\left(\frac{2\pi}{3} - \frac{\sqrt{3}}{4} - \left(\frac{\pi}{2}\right)\right) + \left(6\pi - \left(4\pi - 4\sqrt{3} - \frac{\sqrt{3}}{2}\right)\right)$$

$$A = 12\pi - \frac{9\sqrt{3}}{2} - 9\pi + 2\pi + 4\sqrt{3} + \frac{\sqrt{3}}{2}$$

$$A = 5\pi$$

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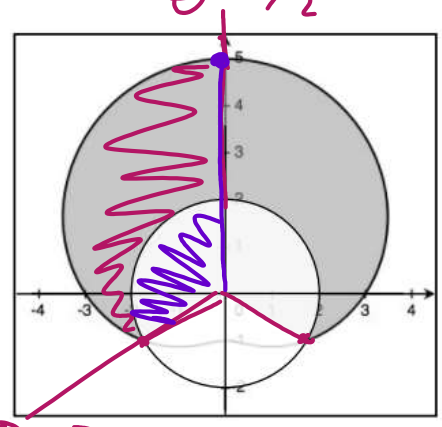
Scan the QR Code above to watch a video covering Example 1

<u>Example 2</u>: Finding the Area of a Region Between Two Curves

Find the area that is inside $r = 3 + 2\sin\theta$ and outside the circle $r = 2$. The graph of these two curves is shown to the right.

$\theta = \pi/2$



$3 + 2\sin\theta = 2$

$2\sin\theta = -1$

$\sin\theta = -\frac{1}{2}$

$\theta = \frac{7\pi}{6}$ & $\frac{11\pi}{6}$

$$A = 2\left[\frac{1}{2}\int_{\pi/2}^{7\pi/6}(3+2\sin\theta)^2\,d\theta - \frac{1}{3}\cdot\pi\cdot 2^2\right]$$

$\theta = \frac{7\pi}{6}$

$\sin^2\theta = \frac{1}{2}(1-\cos 2\theta)$

$$A = \int_{\pi/2}^{7\pi/6}(9 + 12\sin\theta + 4\sin^2\theta)\,d\theta - \frac{8\pi}{3}$$

$$A = \int_{\pi/2}^{7\pi/6}(9 + 12\sin\theta + 2 - 2\cos 2\theta)\,d\theta - \frac{8\pi}{3}$$

$$A = \left[11\theta - 12\cos\theta - \sin 2\theta\right]_{\pi/2}^{7\pi/6} - \frac{8\pi}{3}$$

$$A = 11\left(\frac{7\pi}{6}\right) + \cancel{12}^{6}\left(+\frac{\sqrt{3}}{\cancel{2}}\right) - \frac{\sqrt{3}}{2} - \left(\frac{11\pi}{2}\right) - \frac{8\pi}{3}$$

$$= \frac{77\pi}{6} - \frac{33\pi}{6} + \frac{11\sqrt{3}}{2} - \frac{16\pi}{6}$$

$$= \frac{14\pi}{3} + \frac{11\sqrt{3}}{2} = \frac{28\pi + 33\sqrt{3}}{6}$$

24.187

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